# Deconstructing Credit Risk: A Bayesian Network Approach Analysis of Financial Risk Models.

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**Abstract**

This paper proposes the use of Bayesian Networks (BNs) as a framework for structural interpretation and model auditing in high-dimensional tabular data, moving beyond the traditional "black-box" predictive paradigm. Using the Lending Club dataset as a case study, we rigorously evaluate two distinct structure learning approaches: a Constraint-Based algorithm (PC) and a Search-and-Score algorithm (Hill Climbing with BIC optimization).

The experiments demonstrate that Constraint-Based methods fail to recover the global topology in this domain, resulting in disconnected DAGs due to the vanishing statistical power of high-order conditional independence tests ($G^2$). In contrast, the Search-and-Score approach successfully approximates the joint probability distribution. Topological analysis of the learned graph identifies a critical information bottleneck: the engineered feature 'Loan Grade' acts as a Markov Blanket for the target variable, rendering raw features conditionally independent via d-separation.

By applying structural regularization (node blacklisting and temporal constraints) to mitigate data leakage and remove confounding variables, we derive a refined probabilistic graphical model. This "sanitized" structure reveals that, absent the engineered score, the target class is conditionally dependent on behavioral volatility features rather than static solvency metrics. We conclude that Bayesian Networks provide a necessary methodology for validating feature interaction and detecting structural bias in predictive modeling pipelines.

# 1 Introduction to the Data

## 1.1 Risk Models and Business Context

The dataset used in this work corresponds to the *Lending Club Loan Default Dataset*, publicly available on Kaggle. This dataset contains historical information about personal loans issued through the *Lending Club* platform, an online peer-to-peer lending marketplace in the United States that connects borrowers with individual and institutional investors. The dataset collects data from all its ussers

Figure 1: Lending Club logo.

The platform provides personal loans primarily for debt consolidation, credit card refinancing, and other consumer needs. Having funded over $100 billion in loans since its launch in 2007, and more than $2.6 billion in originations in the third quarter of 2025 alone, the platform has become a major financial intermediary. Furthermore, in accordance with its transparency policy, Lending Club publishes detailed historical loan data, including borrower characteristics and loan outcomes, making it a standard benchmark for credit risk modeling and default prediction research.

It is important to note the structural difference between this peer-to-peer model and traditional banking. Unlike traditional banks, which lend capital from their own balance sheets and are strictly regulated to protect depositors, LendingClub operates as an intermediary marketplace. This allows for a much broader spectrum of borrower profiles, including higher-risk segments that might be excluded from the conservative risk appetite of traditional banking (and therefore higher risk of defaults).

However, despite the fact that the platform is not legally subject to the same regulatory capital constraints as a bank, the risk assessment criteria maintained in this context align with the principles of the *Basel II/III Accords*. To ensure confidence for institutional investors, the underlying risk modeling methodologies specifically the estimation of Probability of Default (PD) adhere to these rigorous industry standards as later in the model will be proved.

## 1.2 Dataset Description

The raw dataset initially comprised 151 variables describing more than 300,000 loan applications over the period of Jul-2007 to Dec-2018. Some of them correspond to internal metrics and evaluations, thus, a feature selection will be done before fitting the networks.

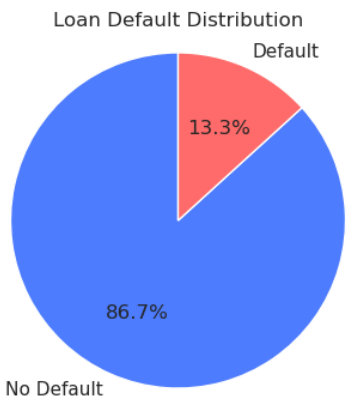


Figure 2: Distribution of var. (`loan_status`).

Each instance in the dataset represents an individual loan application, described by a set of financial and socio-demographic variables. The target information of interest is related to the loan repayment behavior, particularly whether a loan defaults or not. In addition to the target-related variables, the dataset includes features such as loan amount, interest rate, employment length, annual income, debt-to-income ratio, credit history indicators, loan purpose...

The primary target variable for this analysis is the **Loan Status** (fig 2), which indicates the cur-

rent state of the loan. However, the objective of this work is not only to model the probability of default, but also to explore and interpret the dependency relationships among the variables using Bayesian Networks. Focusing on discovering conditional independence relationships and probabilistic dependencies encoded in the data.

## 2 Data Preprocessing

In order to apply Bayesian Network learning techniques, a specific preprocessing pipeline was designed to adapt the original dataset to the Bayesian Network models. The preprocessing steps focus on handling missing values, variable selection, discretization and reducing the number of records.

Initially, only 50,000 observations where randomly selected (which are far enought for this purpose). Then, irrelevant or redundant variables were removed, such as identifiers or attributes with no predictive or interpretative value for probabilistic modeling. Variables with an excessive proportion of missing values were also discarded. For the remaining features, missing values were handled replacing them with the value of 0 in the case of numerical variables or a ”N/A” category in the case of categorical ones.

A key preprocessing step consisted of **discretizing** continuous variables (float) or **merge** into fewer groups those integer or categorical variables with many classes. Since classical Bayesian Networks assume discrete random variables, numerical attributes such as annual income, loan amount, interest rate or debt-to-income ratio were discretized using a quantile-based discretization strategy into 5 gruops. This approach ensures that each discrete interval contains a similar number of observations, which improves the robustness of parameter estimation (not sensible to outliers) and avoids sparsity issues in conditional probability tables.

After preprocessing, the resulting dataset consists exclusively of discrete variables, making it suitable for both structure learning and parameter estimation in Bayesian Networks.

## 3 Feature Subset Selection

Bayesian Networks (BNs) capture complex relationships between variables and result very interesting when displaying them in a directed graph. However including too many variables makes the representation difficult to interpret and thus useless. In this analysis, we employ several techniques to reduce the number of features considered.

The first step in the FS stratetegy was to reduce numerical features, which were seem redundant or are so related that show almost the same patterns. A quick pairwise **Pearson correlation** has been calculated. After, those variables with at least one Pearson correlation over 0.85 have been flagged.

The second step, after discretizating all variables, it has been computed a variable ranking, taking the **Mutual Information (MI)** metric. Best ones where chosen and the rest discarded. If more than one flagged variable appear, the highest one remains and the other leaves. Only 29 variables where chosen (number taken when looking at the decay of the MI score; although it might be still high, it will be taken into consideration later) and approximately 49,500 observations. (More information about the variables on appendix table 10).

# 4 Search-Score Approach Models

Learning the structure of a Bayesian Network from data is, in essence, an optimization problem. The goal is to find a Directed Acyclic Graph (DAG), denoted as $G$, that best represents the joint probability distribution of a dataset $D$.

Score-based methods assign a numerical value to each possible candidate structure, measuring how well the graph fits the data.

Since the number of possible graphs grows super-exponentially with the number of nodes, exhaustive search is infeasible, requiring heuristic algorithms.

## 4.1 Hill Climbing (HC)

The ***Hill Climbing*** **(HC)** algorithm is a greedy local search strategy where, at each iteration, the move that maximizes the score increment is chosen. The process terminates when a local optimum is reached.

Given that a significant number of features remain under study at the end of the FSS, HC appears to be the most suitable approach. Other methods, such as Genetic Algorithms, Simulated Annealing, involve exploring a population of solutions or accepting non-improving moves, which significantly increases computational cost and risk becoming computationally intractable due to the combinatorial explosion (recall that all discretized numerical variables involved have 5 classes which exponentially increase the size of the probability tables).

## 4.2 Scores and Metrics

In this study, the HC strategy has been evaluated using distinct scoring metrics across the complete set of variables to obtain the structure and different parameter estimations.

Specifically, the ***structure learning*** phase has been conducted using a diverse set of scoring functions to evaluate the trade-off between model fit and complexity: **K2**, a widely used Bayesian score; **AIC** (Akaike Information Criterion) and **BIC** (Bayesian Information Criterion), which introduce penalty terms based on the number of parameters to prevent overfitting; and **BDeu** (Bayesian Dirichlet equivalent uniform), which ensures score equivalence for Markov equivalent structures.

Regarding ***parameter estimation***, the study compares **Maximum Likelihood Estimation** (MLE), against Bayesian estimators like **Bayesian Dirichlet** and **BDeu** (Bayesian Dirichlet equivalent uniform). These Bayesian methods mitigate the risk of zero-probabilities in the contingency tables due to data sparsity. However, since the number of observations in this dataset is substantial compared to the number of parameters to be estimated, the smoothing effect of the Bayesian priors becomes negligible. As a result, all three methods converge to nearly identical parameter estimates and therefore MLE will be used since is simpler.

# 5 HC (Hill Climber) + BIC (Bayesian Information Criterion) graph

Based on the comparative analysis of various structure learning algorithms, the network generated using **Hill Climbing with the BIC score** (Bayesian Information Criterion score) has been identified as the best model. It offers the best balance between capturing complex relations of the business and being enough simply to be explainable. Since the number of nodes is significant the best model turns to be the one which penalize the most complexity (less number of arcs). (See the alternative graphs)

The resulting Directed Acyclic Graph (DAG) offers a descriptive view of the data, causal or consequence links (due to use of HC). Although the graph contains a significant amount of information, this section will summarize the findings and focus specifically on the target variable.

**Core Drivers of Default Risk:** The target variable, `binary_loan_status`, is in the center of the network, depending directly on only two key parents. This suggests that, conditional on these two variables, the target is independent of most other ancestors (Markov Blanket property).

- *Grade as a measure of risk:* The variable `grade` acts as the strongest determinant of loan status. As it is shown in the figure 3, it functions as a funnel, it captures all the patterns of the different features and collapse them into a single score according to the probability of default (see table 1 and figure 4). This means that Loan Club platform did a great job. Regarding its parents, it resembles a FICO score, which is used in traditional banking by summarizing the borrower's past financial behavior. As observed in the figure 3, upstream variables such as `bc_util` and `total_bc_limit` (bankcard utilization and limit) or `num_tl_op_past_12m` (total financing ops. in the last year), flow into `grade`, confirming that high utilization and frequent credit seeking negatively impact the assigned grade. Also pointing at `int_rate%` and `term`. (The better paid and the longer duration means riskier).
- *Loan Age, a sign of crisis periods:* The edge `loan_age` captures the temporal dimension. When the loan was funded (in number of months). It is very meaningful, it reflects all expansive and contraction periods of the US economy. 2007-2009 financial crisis and 2014-2015 oil crisis affected severely default rates or loan applications and thus it explains probability of default as well as interest rates (`int_rate%`) (see figure 4). At the same time, this variable is too significant that brings very much the attention of other potential explanatory variables that could point at `grade` instead. Such as `all_util` or `inq_last_12m` (percentage of utilization of all credit lines and inquiries over the last year respectively).

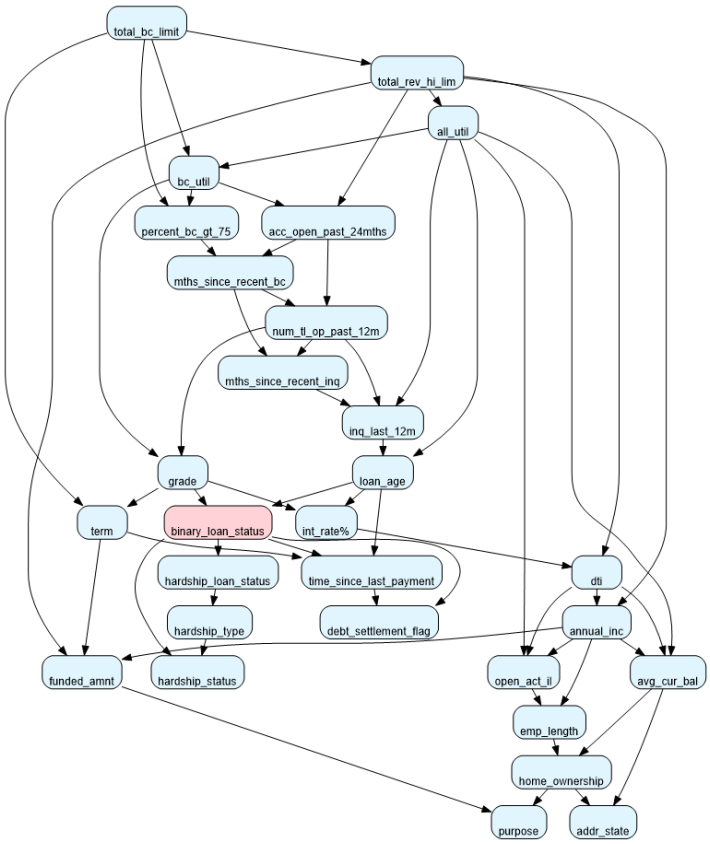


Figure 3: BN graph, HC + BIC using all variables.

**Socio-Economic Interactions.** A distinct cluster in the lower-right section captures the relationship between income, assets (`avg_cur_bal`, `home_ownership`) and employment. At first sight, it seems that `dti` determines socioeconomic factors and they appear not to be *d-connected* with `grade`. However the BN may have other pathways of influence (will be proved later). While `annual_inc` converges into `dti` (representing the debt burden), the asset-related variable `avg_cur_bal` connects directly to `total_rev_hi_lim` in the upper section of the graph. This second path is critical: it suggests that socioeconomic status also determines the credit limits assigned to a borrower. Higher limits (driven by assets) lead to lower utilization ratios (`bc_util` and `all_util`), which are direct parents of `grade`. Therefore, the model successfully captures two dimensions of the borrower's profile: their monthly solvency (via `dti`) and their structural financial capacity (via `total_rev_hi_lim`).

Beyond this relationship, other associations can also be observed, such as the following:

- *Income and Housing:* A logical link is established where `annual_inc` influences `home_ownership`. Higher income levels correlate with a higher probability of having a mortgage or owning a home versus renting.
- *Home Ownership and Purpose:* The edge `home_ownership` and `purpose` reflects lifestyle needs; for instance, homeowners may seek loans for "Home Improvement," while renters might borrow for "House" or "Car" financing.
- *Geography:* The connection `home_ownership` and `addr_state` implies a statistical relationship between housing status and location, likely driven by the varying cost of living and real estate markets across different states (e.g., high ownership rates in rural states vs. high renting rates in metropolitan areas) (see map).

**Structure and Anomalies.** The network exhibits a clear hierarchical flow initiating from root nodes representing credit capacity, specifically `total_bc_limit` and `total_rev_hi_limit`. These variables propagate their influence downstream through utilization metrics (e.g., `bc_util`, i.e. utilization of bankcards) to eventually shape the borrower's risk profile. However, the unconstrained search introduces specific algorithmic artifacts:

- *Causal Inconsistencies:* The model incorrectly positions origination terms (e.g., `term`) as children of the `grade`, contradicting their actual temporal precedence.
- *Data Leakage:* Post-default indicators (e.g., `hardship_status`, i.e. if borrowed is in a hardship plan to facilitate payments, `debt_settlement_flag`) appear as consequences in the graph. These must be pruned, as they reveal the target state rather than predict it.

**Proposed Modifications.** To refine the network for predicting the target, the following expert constraints will be applied:

- *Blacklisting:* Prohibit edges originating from `binary_loan_status` to origination variables (e.g., `term`, `int_rate`, `funded_amnt`) to preserve temporal order.
- *Filtering:* Remove leakage nodes related to debt settlement and hardship plans prior to structure learning. Or nodes which do not appear to have causal relation with the target variable.
- *Expert Edges:* Enforce edges which are assumed to be causally related, such as `grade` to `int_rate`, since the interest rate is explicitly determined by the bank based on the assigned grade.

## 5.1 Conditional Probabilities

By a variable elimination method, $\mathbb{P}(\texttt{loan_status} \mid \texttt{loan_age}, \texttt{grade})$ is obtained from the BN.

| Grade | Jul 07–09 | Oct 09–12 | Jan 12–14 | May 14–16 | Aug 16–18 | Marginal P(Def) |
|---|---|---|---|---|---|---|
| **A** | 4.17% | 4.78% | 5.97% | 5.51% | 1.91% | **3.72%** |
| **B** | 8.82% | 13.25% | 11.29% | 12.27% | 5.84% | **9.08%** |
| **C** | 13.04% | 18.49% | 17.69% | 19.73% | 10.10% | **14.99%** |
| **D** | 33.33% | 25.40% | 21.18% | 28.22% | 14.66% | **21.18%** |
| **E** | 33.33% | 26.53% | 27.27% | 35.64% | 18.71% | **26.94%** |
| **F** | 33.33% | 20.00% | 30.82% | 40.95% | 31.18% | **35.09%** |
| **G** | – | 20.00% | 25.00% | 50.44% | 29.47% | **37.60%** |

Table 1: Default rate by `grade` and `loan_age` calculated from the CPD of the BN graph 3.

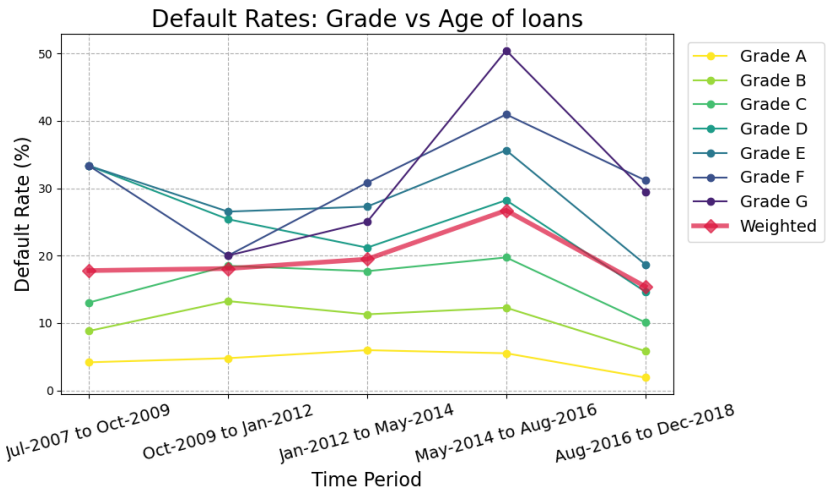


Figure 4: Temporal representation of table 1. Although 2007-2009 period is not well represented by the sample (born of the platform combined with fear and low investment rates), low score loans have greater default rates. On the other hand, it is clear how 2015 crisis inflated default rates at every loan grade. See how it resembles consumers expenditure levels in the USA.

## 5.2 Independencies

### Markov Blankets (MB)

To validate the structural integrity of the Bayesian network, the **Markov Blanket (MB)** of key variables is analyzed. The MB is defined as the minimal set of nodes (comprising parents, children, and spouses) that renders a specific variable conditionally independent from the rest of the network ($X \perp Y \mid MB(X))\,(\forall Y \notin MB(X))$. Here are presented a set of MB for some relevant variables which are not trivial (have children and spouses):

1. **MB of `binary_loan_status`:** *Variables:* {`grade`, `loan_age`, `debt_settlement_flag`, `hardship_status`}.
   Its blanket is contaminated by obvious *leakage* (`debt_settlement_flag` and `hardship_status`), which are future events only available when default and they will be removed in the next BN model. Structurally, `grade` acts as a *blocking mechanism*: once the node knows the `grade`, variables like `annual_inc` or `dti` become conditionally independent. Therefore it will be removed too.

2. **MB of `grade` (The Assessment):** *Variables:* {`int_rate%`, `bc_util`, `binary_loan_status`, `loan_age`, `num_tl_op_past_12m`, `total_bc_limit`, `term`}.
   It is the central information node. Its parents are the raw history data, and its function is to compress that financial biography into a single metric. Its MB indicates that, given the parents, any other past history is ignored. In turn, `grade` is the dominant parent of both price and outcome, orchestrating the network.

3. **MB of `avg_cur_bal` (The Wealth Proxy):** *Variables:* {`all_util`, `home_ownership`, `dti`, `addr_state`, `emp_length`, `annual_inc`}.
   This blanket represents the borrower's *Accumulated Wealth and Financial Footprint*. Unlike income (which is cash flow), `avg_cur_bal` acts as a indicator also for asset "stock". It is structurally driven by socioeconomic markers: `home_ownership` (mortgage holders naturally have higher account balances) and `annual_inc` (capacity to accumulate). `addr_state` acts as a necessary context for cost-of-living adjustments, while `emp_length` signals the stability required to build these balances over time.

## 5.3 Inference Validation: "What-If" Scenarios

Once the network structure (DAG) is established, exact inference is used via *Variable Elimination* to validate whether theoretical conditional independencies hold true in real data. Three questions are proposed:

**Question A: Conditional Independence Test of `grade` (The "Funnel" Effect):**

It is observed that the probability of default barely shifts when introducing information about debt-to-income ratio (`dti`) once the `grade` is known according to the BN. This means that the Lending Club model achieves high accuracy when assigning scores to borrowers.

- **Hypothesis:** Given a specific `grade`, knowing the debt level (`dti`) should not provide significant additional information regarding default probability $(\text{Default} \perp \text{dti} \mid \text{Grade})$.

$$P(\text{Default} \mid \text{Grade} = X) \approx P(\text{Default} \mid \text{Grade} = X, \text{dti} = \text{High})$$

- **Result:** The experiment was repeated for five different grades representing the full range of credit quality.

| Scenario (Grade) | Base P(No Default) | With DTI=High | $\Delta$ (Impact) |
|---|---|---|---|
| Grade A (Low Risk) | 0.9628 | 0.9631 | **+0.0003** |
| Grade B | 0.9092 | 0.9099 | **+0.0007** |
| Grade C | 0.8501 | 0.8513 | **+0.0012** |
| Grade D | 0.7882 | 0.7920 | **+0.0038** |
| Grade E | 0.7306 | 0.7317 | **+0.0011** |
| Grade F (High Risk) | 0.6491 | 0.6498 | **+0.0007** |

Table 2: Test of Grade blocking effect on DTI across Grades A-F.

- **Interpretation: Hypothesis Confirmed Robustly.** The variation is consistently negligible (max $\Delta \approx 0.38\%$) across the entire spectrum. Whether the client is prime (Grade A, 96% success) or subprime (Grade F, 64% success), adding `dti` information does not shift the prediction once the `grade` is fixed. This mathematically proves that the `grade` node effectively encapsulates the risk information from its parents. However there is a tiny structural bias that could be fixed on the rating model, it systematically overestimates `dti` value.

**Question B: Sensitivity to Term:**

Does the variable `term` adds predictive power beyond the `grade` across different risk profiles?

- **Hypothesis:** A longer loan term implies a longer exposure to risk. Probability of success should decrease ($P(\text{Success} \mid 60m) < P(\text{Success} \mid 36m)$). Does `grade` consider that?

$$P(\text{Target=1} \mid \text{Gr} = X, \text{Term} = 60m) < P(\text{Target=1} \mid \text{Gr} = X, \text{Term} = 36m)$$

- **Result:** The experiment was conducted across four grades ranging from low risk (B) to extreme risk (G).

| Grade Profile | P(Target=1) 36m | P(Target=1) 60m | $\Delta$ (Impact) |
|---|---|---|---|
| Grade B (Low Risk) | 0.9090 | 0.9100 | **+0.0010** |
| Grade D (Med Risk) | 0.7877 | 0.7887 | **+0.0010** |
| Grade F (High Risk) | 0.6478 | 0.6495 | **+0.0017** |
| Grade G (Extreme) | 0.6242 | 0.6239 | **-0.0003** |

Table 3: Impact of loan term on success probability conditioned on Grade.

- **Interpretation: Hypothesis Refuted Systemically.** The model shows near-total insensitivity to the loan term across the entire risk spectrum. The variations are negligible (avg $\Delta \approx 0.1\%$) and inconsistent (sometimes positive, sometimes negative). This confirms that in the learned BN structure, the `grade` node excessively dominates the prediction, rendering the `term` variable conditionally independent of the outcome once the rating is fixed.

**Question C: Sensitivity to Loan Purpose:**

Does the variable `purpose` discriminate risk significantly when the loan amount is fixed?

- **Hypothesis:** Different loan purposes carry inherent risks. While taking a loan for home improvement or education may be considered an *investment* (implying future stability), borrowing for a vacation, wedding, or car is considered *consumption* with no expected financial return. Therefore, we expect a significant spread in success probability between investment-driven and consumption-driven purposes.

$$P(\text{Success} \mid \text{Amt} = X, \text{Purp} = \text{Inv}) > P(\text{Success} \mid \text{Amt} = X, \text{Purp} = \text{Cons})$$

- **Result:** The experiment compared the highest and lowest probability purposes across five loan amount bins.

| **Loan Amount** | **Best Purpose** | **Worst Purpose** | **$\Delta$ (Gap)** |
|---|---|---|---|
| 1k - 7k (Low) | educational | car | **1.27%** |
| 7k - 10.5k (Med-Low) | home_improvement | educational | **0.58%** |
| 10.5k - 15.1k (Med-High) | educational | wedding | **1.81%** |
| 15.1k - 23.3k (High) | renewable_energy | wedding-car | **2.82%** |
| 23.3k - 40k (Very High) | renewable_energy | debt_consolidation | **1.32%** |

Table 4: Risk spread between best and worst loan purposes by amount.

- **Interpretation: Hypothesis Partially Refuted / Low Sensitivity.** Statistically, the `purpose` shows limited predictive power once the loan amount is fixed, with a maximum observed probability gap of only $2.82$% (in the High amount bin). The variable is largely blocked by other factors.

  However, the **qualitative ranking supports the investment logic**:

- **Consumption is risky:** Pure consumption categories like `car` and `wedding` consistently appear as the worst performers across multiple bins.

  For the very highest loans (23k-40k), `debt_consolidation` becomes the worst purpose, likely signaling borrowers in severe financial distress trying to restructure liabilities.

- **Investment is saver:** For high loan amounts ($>$15k), `renewable_energy` emerges as the safest purpose, likely correlating with home ownership and higher disposable income but also with an investment to decrease future bills. The same goes for `educational` proposes, which may imply higher salaries or more stability.

# 6 Structural Refinement and Validation

To address the structural inconsistencies and data leakage identified in the initial analysis, a refined model was developed to uncover the true financial causalities obscured by the bank's scoring system. By imposing strict expert constraints, this approach eliminates post-default variables (e.g., *hardship_status*), which are not of interest for this study, and enforces temporal consistency to prevent reverse causality. Crucially, the *grade* variable was explicitly blacklisted to remove its "funnel effect," forcing the network to learn the direct probabilistic dependencies between raw financial metrics and loan default instead of directing them to the score variable.

## 6.1 Analysis of the Refined Structure

The resulting graph (Figure 5) reveals the true structural drivers of default risk when the artificial score is absent.

**1. Restoration of Temporal Logic:** Unlike the initial model, the algorithm now correctly identifies *term* (36 vs. 60 months) as a direct parent of *binary_loan_status*. This confirms the financial axiom that longer exposure time increases default risk. Furthermore, a logical business rule was captured: *funded_amnt* $\rightarrow$ *term*, indicating that higher loan amounts drive the decision to extend the loan duration to lower monthly payments.

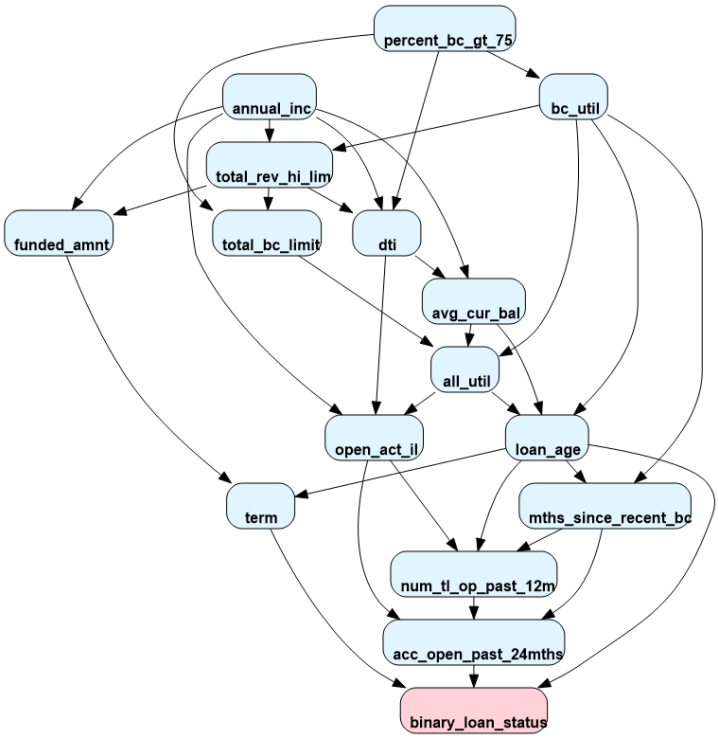


Figure 5: Refined Bayesian Network structure (HC + BIC) applying expert constrains.

**2. The New Markov Blanket of Default:** Without the *grade* variable, the Markov Blanket of the target variable shifts to describe a "Volatility vs. Stability" profile. The direct parents of the target are now:

- **term:** The structural risk of the contract.
- **loan_age:** Encodes the time when the loan was granted.
- **acc_open_past_24mths** (Accounts opened in past 24 months): This node becomes a critical predictor. It represents the borrower's recent "hunger for credit." A high velocity of opening new accounts signals financial distress or aggressive leveraging.
- **mths_since_recent_bc:** The recency of the last bankcard opening. Together with the previous variable, it measures behavioral volatility.

**3. The Solvency Hierarchy:** The upper part of the graph forms a "Solvency Tree" rooted in *annual_inc*. The network correctly learns that income determines credit limits (*total_rev_hi_lim*, *total_bc_limit*), which determines utilization ratios (*dti*, *bc_util*). Crucially, these solvency metrics (Income, DTI) are **not** direct parents of the target in this graph. This suggests that while solvency defines the *capacity* to borrow, it is the *recent credit-seeking behavior* (volatility) that triggers the actual default. This aligns closely with *Basel II* and traditional banks' rating systems.

# 7 Probabilistic Model (PC Algorithm)

In addition to the score-based approaches detailed in previous sections, constraint-based methods, specifically the **PC (Peter-Clark)** algorithm, were evaluated. However, this approach was ultimately discarded as the resulting graphs were largely uninformative and frequently disconnected. Even when relaxing the significance level ($\alpha$) to encourage edge inclusion, the algorithm tended to identify relationships that were consequential rather than causal, thereby failing to capture the true underlying drivers of the credit risk model.

# 8 Approximate inference

The approximate inference does not significantly alter the marginal distributions; therefore, the analogous tables have been omitted to avoid redundancy. Testing with 5,000 cases (10% of the dataset) revealed that the estimates remained highly stable, with a maximum difference of approximately 4% compared to the exact inference in all of the above cases.

# 9 Conclusions

This study has explored the application of Bayesian Networks to the Lending Club dataset, moving beyond simple classification to structural auditing of financial risk. The key findings are summarized below:

- **The "Grade" as a Sufficient Statistic:** The comparison between the initial model and the refined model demonstrates the efficiency of Lending Club's rating system. The variable *grade* acts as a near-perfect information funnel, absorbing the variance of financial history and rendering raw variables conditionally independent of the outcome. When *grade* is present, it is the sole necessary predictor; when it is removed, the network must reconstruct risk through behavioral proxies.
- **Behavior Trumps Solvency in Default Prediction:** The structural analysis reveals that default is not driven linearly by income or debt-to-income (DTI) ratios. Instead, the "Refined Model" shows that risk is structurally determined by the loan characteristics (term) and the borrower's recent volatility (opening multiple accounts in a short window). High-income borrowers with aggressive recent credit behavior are riskier than stable low-income borrowers.
- **Bayesian Networks as Auditing Tools:** Beyond their predictive capacity, BNs proved invaluable for interpretability. They successfully identified data leakage in the dataset (post-default variables) and allowed us to visualize the causal chain from income to credit limits. This transparency is a distinct advantage over "black-box" models like Neural Networks, especially in regulated financial environments where explainability is mandatory.

# 10 Appendix 1: Variable description

| Variable | Type | Description |
|---|---|---|
| acc_open_past_24mths | Integer | Number of trades opened in the past 24 months. |
| all_util | Float | Balance to credit limit ratio on all trades. |
| annual_inc | Float | Self-reported annual income provided by the borrower during registration. |
| avg_cur_bal | Float | Average current balance of all accounts. |
| bc_util | Float | Ratio of total current balance to high credit/credit limit for all bankcard accounts. |
| dti | Float | Debt-to-Income ratio calculated using total monthly debt payments divided by monthly income. |
| empl_length | Integer | Number of years employed in the current position. |
| inq_last_12m | Integer | Number of credit inquiries in the past 12 months. |
| mths_since_recent_bc | Integer | Months since the most recent bankcard account was opened. |
| mths_since_recent_inq | Integer | Months since the most recent credit inquiry. |
| num_tl_op_past_12m | Integer | Number of accounts opened in the past 12 months. |
| open_act_il | Integer | Number of currently active installment trades. |
| open_rv_24m | Integer | Number of revolving trades opened in the past 24 months. |
| percent_bc_gt_75 | Float | Percentage of all bankcard accounts with a balance greater than 75% of the limit. |
| total_bc_limit | Float | Total bankcard high credit/credit limit. |
| total_rev_hi_lim | Float | Total revolving high credit/credit limit. |
| int_rate% | Float | Interest Rate on the loan. |
| funded_amnt | Float | Amount funded by investors. Can be lower than the amount asked by the borrower (loan_amount). |
| loan_age | Integer | Derived variable representing the age of the loan in months (time since origination). |
| time_since_last_pymnt | Integer | Derived variable representing the time elapsed since the last payment was received. |
| addr_state | Nominal | Categorical variable indicating the borrower's geographic region (e.g., North, South, West). |
| grade | Ordinal | LC assigned loan grade reflecting credit quality. |
| home_ownership | Nominal | Home ownership status provided by the borrower (e.g., RENT, OWN, MORTGAGE). |
| purpose | Nominal | Category provided by the borrower describing the reason for the loan request. |
| term | Binary | The number of payments on the loan (typically 36 or 60 months). |
| hardship_type | Nominal | Describes the type of hardship plan offering. |
| hardship_status | Nominal | Describes if the hardship plan is active, pending, canceled, completed, or broken. |
| hardship_loan_status | Nominal | Loan Status as of the hardship plan start date. |
| debt_settlement_flag | Binary | Flags whether the charged-off borrower is working with a debt-settlement company. |
| binary_loan_status | Binary | Derived target variable indicating if the loan was fully paid (0) or defaulted/charged-off (1). |

Table 5: Variables from the dataset selected by a FSS analysis. Name, type and brief description.

## References

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